

ISBI 2027 submission and evidence audit

3 October 2026. This audit distinguishes measured evidence from claims that remain unsupported. No review process or acceptance can be guaranteed.

Submission decision

The strongest defensible paper is a controlled empirical study of edge priors and domain shift, not a claim of a new state-of-the-art breast-cancer detector. The work is 2D semantic nuclear foreground segmentation. BBBC039 is U2OS fluorescence, not breast tissue; TNBC is breast histology used as a zero-shot stress test. Preliminary 3D work is mentioned only as future direction.

The prepared manuscript uses the official ISBI-linked IEEE style, is four pages, and places all technical content on pages 1–3. Page 4 contains only compliance, acknowledgments/disclosures and references. ISBI 2027 is single-blind; the current author names are therefore present. The official deadline shown on the ISBI author page is 26 October 2026 at 11:59 PM US Eastern time. Recheck the portal before submission: <https://biomedicalimaging.org/2027/papers/>.

What was newly tested

Evidence block	Runs / units	Principal result	Interpretation
Original public component study	7 models x 3 seeds, 420 epochs	Gabor bundle 96.85% Dice vs vanilla B7 96.60%; +0.250 points, 95% CI +0.197 to +0.299	Small within-screen gain
Matched architecture baselines	6 models x 3 seeds, 360 epochs	UNet++/Xception/SCSE 96.87%; Gabor minus baseline -0.014 points, CI -0.047 to +0.018, Holm $p=.378$	No evidence that Gabor beats the strongest matched comparator
Validation-only sensitivity	10 conditions x 3 seeds, 600 epochs	96.541–96.731% foreground Dice across filter/sampler choices	Gabor settings are not uniquely optimal; effects are small
TNBC explicit grayscale audit	21 checkpoints x 50 images	RGB-entry and explicit-grayscale features match exactly; 115/115 TNBC+BBBC038 images pass	The low transfer score was not caused by omitted grayscale conversion
TNBC one-shot polarity stress test	21 checkpoints, 11 patient groups	Vanilla 8.07% to 51.30% patient-macro Dice; +43.24 points, patient bootstrap CI +36.55 to +50.15	Bright/dark polarity explains much, but not all, domain failure

Total new controlled training: 69 runs and 1,380 fixed-budget epochs. Every run, including weak models, is retained.

External literature comparison

These figures are context, not a common leaderboard. Metric definitions, splits, instance postprocessing and model-selection practices differ.

Study	Dataset/protocol	Reported result	Comparability to this work
Caicedo et al., Cytometry A (2019)	BBBC039 official split; object F1 averaged over IoU 0.50–0.90	U-Net 0.898; DeepCell 0.858	Not semantic Dice; useful historical context
SegmentGraph, IJCAI (2022)	BBBC039 official split	pixel Dice 0.9482; AJI 0.8680; object F1 0.9670	Pixel Dice is closer, but graph postprocessing and instance endpoints differ
ASW-Net (2022)	BBBC039 official split with interior expansion	DICE1 0.96452; DICE2 0.84798; AJI 0.90200	DICE1 construction differs from native semantic Dice
MA-Net, PLOS ONE (2024)	BBBC039	Dice 0.973; TNBC 0.809	Public description does not establish a training/selection protocol identical to ours

Study	Dataset/protocol	Reported result	Comparability to this work
NUSeg / pi-PhenoDrug (2024)	BBBC039 fivefold CV	Dice 0.9693; IoU 0.9406	Paper describes per-epoch test evaluation and best-weight selection; not a held-out equivalent
CellTranspose, WACV (2023)	BBBC039 to selected TNBC tissue types with 3/5/10-shot adaptation	pixel-F1 0.6702/0.7377/0.7568; Cellpose 0.5829	Different subset and adaptation setting; our 0.5130 is all-50 zero-shot after fixed inversion

Primary sources: BBBC039 provider <https://bbbc.broadinstitute.org/BBBC039> ; BBBC038 provider <https://bbbc.broadinstitute.org/BBBC038> ; TNBC v1.1 <https://zenodo.org/records/2579118> ; Caicedo evaluation <https://pmc.ncbi.nlm.nih.gov/articles/PMC6771982/> ; SegmentGraph <https://www.ijcai.org/proceedings/2022/0169.pdf> ; ASW-Net <https://pmc.ncbi.nlm.nih.gov/articles/PMC8950038/> ; MA-Net <https://doi.org/10.1371/journal.pone.0308326> ; NUSeg <https://doi.org/10.1002/aisy.202400635> ; CellTranspose <https://pmc.ncbi.nlm.nih.gov/articles/PMC10760785/> .

Reviewer-comment closure

Resolved computationally in the new paper: empty-mask definitions and strata; official split and SHA-256 leakage checks; public code structure; removal of the incorrect Tversky recall claim; explicit boundary loss and BF1/surface-Dice/HD95; orientation/frequency/sigma/Sobel/neutral-edge sensitivity; equal budgets; explicit sampler formula and ablation; safe zero-positive batches; removal of clinical-grade claims; paired bootstrap intervals and Holm-adjusted tests; real-data figures and AI disclosure; TNBC transfer; identical no-TTA evaluation; three seeds and complete learning curves; B7/encoder/architecture controls; no hidden crop generation; recomputed consistent metrics; and retention of failed baselines.

Partially resolved or inherently unavailable: public provider annotations replace undocumented private labels, but this does not create inter-annotator agreement; BBBC039 is one acquisition study, not biological multi-site replication; TNBC inversion was motivated and evaluated post hoc, so it needs confirmation on unseen histology; BBBC039 partly overlaps BBBC038; foundation-model pretraining overlap is unresolved; binary masks do not permit instance split/merge validation; three seeds do not characterize all training variation.

Submission blockers and risk-ranked next experiments

1. **Ethics determination — blocking.** A ready-to-submit UWM determination request is included in submission/UWM_IRB_DETERMINATION_REQUEST.md. Obtain the formal non-human-subjects/exempt determination for reused TNBC human histology and insert its exact wording/identifier. Do not infer ethics status from CC BY licensing. 2. **Immutable reproducibility archive — partial.** The repository is public. Settle inherited-code licensing, upload the selected checkpoint to a GitHub release/Zenodo archive, and cite an immutable DOI or tagged commit. 3. **Unseen histology confirmation — highest scientific value.** Freeze the polarity rule without further TNBC label use; evaluate once on a patient-separated external cohort with compatible nuclear masks. Report patient bootstrap CIs and predeclare the primary endpoint. 4. **Patient-separated histology adaptation.** Compare zero-shot, stain normalization, polarity canonicalization, and limited-shot fine-tuning under grouped folds. Apply all preprocessing identically to every baseline. 5. **Instance evidence.** If the paper claims touching-nucleus separation, train/evaluate against instance IDs using AJI, PQ, object F1 and split/merge errors. Current binary foreground results cannot support that claim. 6. **Compute reporting.** Add per-model training time, inference throughput and GPU memory from the retained summaries if the final paper emphasizes efficiency. Parameter counts alone are insufficient. 7. **Independent seed replication.** Five seeds would narrow optimization uncertainty, but unseen acquisition-level validation is more important than adding seeds to the same chemical screen.

Package integrity

The Overleaf ZIP contains main.tex, references.bib, official style files, actual-data figures, README and checklist. The code ZIP excludes raw data, predictions and 14 GB of checkpoints; it includes source, tests, locked configs, scripts, per-image CSVs, histories, manifests and manuscript/audit sources. The separate dataset ZIP contains provider archives with attribution, licenses/source links and checksums. Checkpoints should be distributed separately through a versioned release because individual B7 files exceed normal

GitHub file limits.